

Grover for satisfiability problems

Exercise 1 - 3-SAT Problem

The 3-Satisfiability Problem (3-SAT) is a classic problem in computational complexity theory and an important example of an NP-complete problem. It asks whether an assignment of boolean variables exists that satisfies a given boolean formula. The formula is expressed in conjunctive normal form (CNF), where each clause contains exactly three literals. A literal is a boolean variable or its negation, and a clause is a disjunction (logical OR) of literals. The entire formula is a conjunction (logical AND) of these clauses. Consider the Boolean formula below:

$$F = (x_1 \vee x_2 \vee x_3) \wedge (\neg x_1 \vee \neg x_2 \vee x_3) \wedge (x_1 \vee \neg x_2 \vee \neg x_3)$$

In this formula, each clause is enclosed in parentheses and contains exactly three literals combined using the logical OR operator $\vee$. The entire formula F is a conjunction of these clauses combined using the logical AND operator $\wedge$.

The goal is to determine whether an assignment of truth values (true or false) exists to the variables x_1, x_2, x_3 such that the entire formula F evaluates to true. This involves checking all possible combinations of truth values for the variables and identifying those that satisfy all the clauses simultaneously.

Tasks:

- 1) Propose a solvable 3-SAT boolean formula.
- 2) Implement Grover's algorithm to find a solution.
- 3) Assess the algorithm's efficiency in finding a solution.

Exercise 2 - 2x2 binary sudoku

Sudoku is a popular logic-based number-placement puzzle. The objective is to fill a grid with numbers so that each row, column, and designated subgrid contains all numbers without repetition. For instance, in a 4x4 grid, there are 16 values to be assigned, with each position in the grid taking values 0 to 15. In a 2x2 sudoku, the problem is much simpler, and the rules are reduced to:

- 1) No column may contain the same value twice
- 2) No row may contain the same value twice
- 3) Assignments are binary.

As you should convince yourself, there are only two possible binary assignments to solve the problem.

Tasks:

- 1) Implement Grover's algorithm to find a solution to the problem.
- 2) Assess the algorithm's efficiency and complexity.
- 3) Propose a generalization for 3x3 sudoku and discuss potential limitations and resources required.